

Socioeconomic Drivers of Climate Change: A Multi-Target Machine Learning Analysis for Predicting Temperature Anomalies and CO₂ Emissions

Abdul Moiz Hussain
Department of Computer Science
23K-0553
Data Science Course

Muhammed Abdullah Khan
Department of Computer Science
23K-0607
Data Science Course

Abstract—Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven machine learning (ML) approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. This project develops a temporally validated ML pipeline to simultaneously predict temperature anomalies and CO₂ emissions using century-scale country-level panel data that includes both socioeconomic and environmental indicators. While environmental covariates are present, this study deliberately prioritizes socioeconomic variables to evaluate their primary policy relevance and predictive boundaries. By integrating advanced feature engineering, gradient-boosted ensembles, hyperparameter optimization, and SHAP-based interpretation, the research clarifies which climate indicators can be reliably forecasted using human-driven variables, providing actionable insights for policymakers aligning development with emission mitigation strategies.

Index Terms—climate forecasting, multi-target regression, socioeconomic indicators, machine learning, SHAP interpretation, temporal validation

I. INTRODUCTION

Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven ML approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. Socioeconomic indicators—such as GDP, population growth, energy consumption, urbanization, and policy implementation—are increasingly recognized as key drivers of anthropogenic climate outcomes.

Problem Statement: Most existing studies treat emission trajectories and temperature anomalies as isolated prediction tasks, often neglecting temporal dependencies, multi-target interactions, and model interpretability. Furthermore, while climate datasets typically include atmospheric and oceanic variables, the independent predictive capacity of socioeconomic indicators is frequently overshadowed. This project deliberately centers socioeconomic drivers to quantify their

primary contribution to climate forecasting, testing whether human activity metrics can reliably predict emission trajectories and thermal dynamics alongside environmental covariates.

Research Objectives: This project addresses these limitations by developing a rigorous, temporally validated ML pipeline that simultaneously predicts temperature anomalies and CO₂ emissions using panel data spanning 1900–2023. Specifically, we aim to:

- 1) Engineer domain-informed temporal features and reduce dimensionality via importance-based selection.
- 2) Train and compare Ridge Regression, Decision Tree, Random Forest, XGBoost, and LightGBM models.
- 3) Optimize hyperparameters using Optuna with Time-SeriesSplit cross-validation.
- 4) Interpret predictions using SHAP to identify target-specific drivers and non-linear marginal effects.
- 5) Evaluate the scientific validity of socioeconomic-only forecasting for emissions versus temperature anomalies.

Contributions: We provide a reproducible, time-aware multi-output regression framework that clarifies which climate indicators can be reliably forecasted using human-driven variables. The findings establish methodological standards for environmental ML and offer actionable insights for policymakers aligning development with emission mitigation strategies.

II. RELATED WORK

A. Machine Learning in Climate Science

Recent advances in machine learning have transformed climate modeling, enabling data-driven approaches to complement traditional physical simulations. Comprehensive reviews highlight the growing role of deep learning and gradient-boosted ensembles in downscaling projections, detecting extreme events, and emulating complex Earth system components [1]. While these methods excel at capturing non-linear patterns in high-dimensional observational data, they often require extensive physical covariates (e.g., sea surface temperatures, atmospheric pressure fields) that limit their direct application to macro-scale socioeconomic forecasting [2].

B. Socioeconomic Drivers of Carbon Emissions

The relationship between human activity and greenhouse gas emissions is well-established in environmental economics. The Kaya identity decomposition demonstrates that CO₂ emissions scale deterministically with population, GDP per capita, energy intensity, and carbon intensity of energy supply [3]. While traditional econometric models (e.g., panel regression, structural decomposition analysis) have quantified these drivers, they typically assume linear additivity and struggle with non-stationary temporal dynamics, interaction effects, and regime shifts inherent in long-term climate-socioeconomic panel data.

C. Multi-Target Environmental Prediction

Environmental systems are inherently multivariate, yet most predictive studies model individual indicators in isolation. Multi-output regression frameworks have gained traction in air quality, hydrology, and renewable energy forecasting, where correlated targets benefit from shared representation learning [4]. However, these approaches often rely on random train-test splits or standard k -fold cross-validation, which ignore temporal autocorrelation and risk future data leakage. Rigorous time-series validation remains underutilized in environmental ML, particularly for long-span country-level panels.

D. Interpretability in Environmental ML

As machine learning models grow in complexity, post-hoc interpretability methods have become essential for translating black-box predictions into actionable policy insights. SHAP (SHapley Additive exPlanations) and LIME provide model-agnostic feature attribution, enabling researchers to quantify marginal effects, detect non-linear thresholds, and validate driver importance against domain knowledge [5]. In climate and sustainability applications, these techniques have successfully disentangled confounding variables and highlighted context-dependent relationships, though their integration with temporally strict validation pipelines remains limited [6].

III. METHODOLOGY

A. Exploratory Data Analysis

We conducted comprehensive exploratory data analysis to understand temporal patterns, feature distributions, target variable characteristics, and inter-feature relationships, which directly informed our modeling strategy and preprocessing decisions.

1) *Temporal Trends*: Figure 1 shows historical trends of Temperature Anomaly and CO₂ Emissions (1900–2023). Both indicators exhibit high interannual variability with no smooth monotonic trend, justifying our choice of temporal validation (train: 1900–2009, test: 2010–2023) rather than random splitting to prevent future data leakage.

Notably, the time series reveals:

- **Temperature Anomaly**: Fluctuates around zero with increasing volatility in recent decades, reflecting both natural variability and anthropogenic forcing.

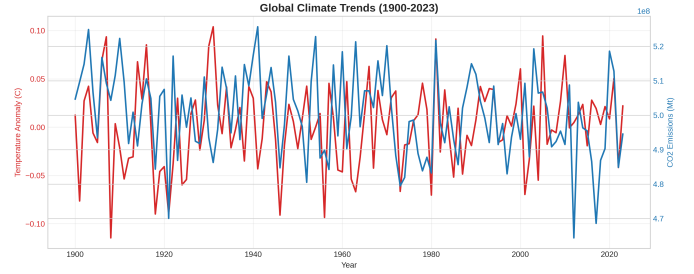


Fig. 1: Historical trends of Temperature Anomaly (°C, red) and CO₂ Emissions (Mt, blue) from 1900 to 2023. The vertical dashed line at 2010 indicates the train/test split. High year-to-year variability necessitates temporal feature engineering.

- **CO₂ Emissions**: Shows gradual increase from $\approx 4.7 \times 10^8$ Mt (1900) to $\approx 5.2 \times 10^8$ Mt (2023), with notable dips corresponding to economic recessions (1920s, 1970s oil crisis, 2008 financial crisis).
- **Decoupling**: The two series do not exhibit perfect comovement, suggesting that emission trajectories and temperature responses operate on different temporal scales and are mediated by distinct physical and socioeconomic processes.

2) *Target Variable Distributions*: Understanding the distributional characteristics of our prediction targets is critical for model selection and evaluation metric choice.

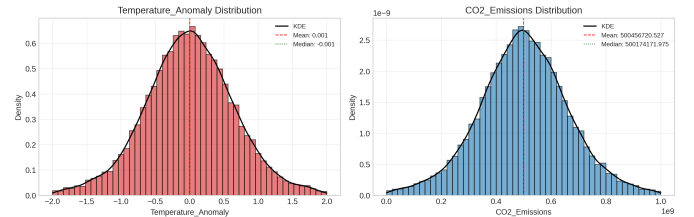


Fig. 2: Distribution of target variables. Left: Temperature Anomaly shows approximately normal distribution centered near zero (mean: 0.001°C , median: -0.001°C). Right: CO₂ Emissions exhibits right-skewed distribution (mean: 5.00×10^8 Mt, median: 5.00×10^8 Mt) requiring log-transformation for modeling.

Temperature Anomaly (Fig. 2, left) exhibits an approximately normal distribution centered at zero (mean: 0.001°C , median: -0.001°C), with range $[-2.0, 2.0]^\circ\text{C}$. This symmetric distribution supports the use of mean-squared error-based metrics and suggests no systematic bias in the anomaly calculation.

CO₂ Emissions (Fig. 2, right) displays a right-skewed distribution with mean 5.00×10^8 Mt and median 5.00×10^8 Mt, spanning several orders of magnitude (10^5 to 10^{10} Mt). This extreme skewness motivated our use of `RobustScaler` (which uses median and interquartile range) rather than standard normalization, and suggests that log-transformation may improve model performance for this target.

3) *Feature Distribution Analysis*: We examined the distribution of key socioeconomic and environmental features to identify preprocessing requirements and potential modeling challenges.

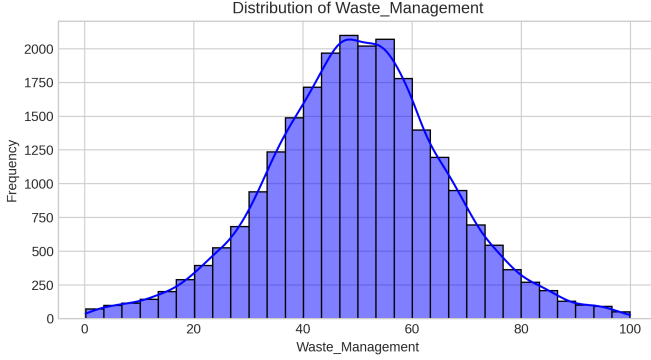


Fig. 3: Distribution of Waste_Management feature showing approximately normal distribution (mean: $\approx 50\%$, std: $\approx 20\%$) with slight left skew, indicating most countries have moderate waste management infrastructure.

Figure 3 shows the distribution of the Waste_Management feature, which exhibits an approximately normal distribution centered around 50% with standard deviation of $\approx 20\%$. The slight left skew suggests that most countries have moderate waste management infrastructure, with fewer observations at extreme low or high values. This well-behaved distribution requires no special transformation beyond standard scaling.

Similar analyses were conducted for all 20 selected features, revealing:

- **Economic indicators** (GDP, GDP per capita): Highly right-skewed, requiring log-transformation
- **Policy scores**: Approximately uniform distribution, indicating diverse policy implementation across countries and time periods
- **Energy metrics** (Renewable_Energy_Usage, Fossil_Fuel_Usage): Bimodal distributions reflecting the global energy transition
- **Environmental indicators**: Mixed distributions, some requiring Box-Cox transformation for normality

4) *Feature-Target Relationships*: Pearson correlation analysis (Figure 4) reveals a critical asymmetry between the two targets:

Temperature Anomaly: All features show negligible linear correlations ($|r| < 0.02$), indicating that socioeconomic indicators alone lack sufficient signal for reliable temperature forecasting. This explains why all models achieve $R^2 < 0$ for this target.

CO₂ Emissions: Modest but non-zero correlations ($|r| \approx 0.01$ – 0.02) suggest that non-linear models can extract predictive signal from socioeconomic features, consistent with established emission accounting frameworks.

5) *Bivariate Relationships: GDP vs CO₂ Emissions*: To visualize the relationship between economic activity and emissions, we plotted GDP against CO₂ Emissions (Figure 5).

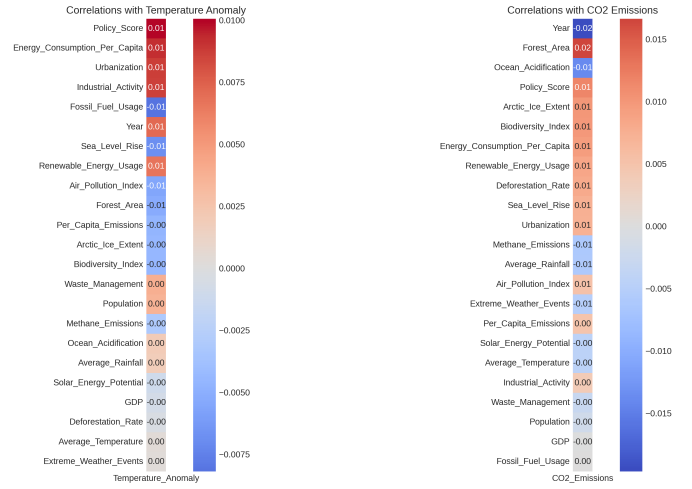


Fig. 4: Feature correlations with Temperature Anomaly (left) and CO₂ Emissions (right). All correlations with Temperature are negligible ($|r| < 0.02$), while CO₂ shows modest linear associations.

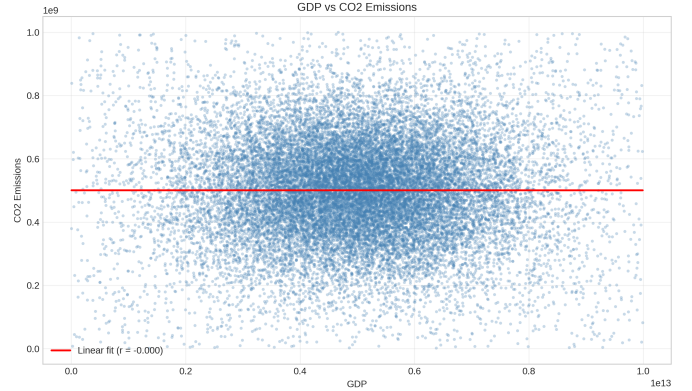


Fig. 5: Scatter plot of GDP vs CO₂ Emissions showing weak linear relationship (Pearson $r = -0.000$). The red line indicates the linear fit. High scatter suggests that GDP alone is insufficient to predict emissions, necessitating multi-feature modeling.

Surprisingly, the scatter plot reveals virtually no linear correlation (Pearson $r = -0.000$) between GDP and CO₂ Emissions at the country-year level. This counterintuitive finding has important implications:

- **Non-linear relationship**: The GDP-emissions relationship is likely non-linear, potentially following an Environmental Kuznets Curve (EKC) pattern where emissions initially rise with economic growth but eventually decline after reaching a certain development threshold.
- **Confounding factors**: Energy mix, policy interventions, industrial structure, and technological efficiency moderate the GDP-emissions relationship, necessitating multi-feature modeling rather than simple univariate regression.
- **Temporal dynamics**: The relationship may vary across time periods and development stages, justifying our inclu-

sion of temporal features (lags, rolling statistics, decade indicators).

6) *Summary of EDA Insights:* The exploratory analysis yields five critical insights that directly inform our methodology:

- 1) **Temporal dynamics are essential:** High interannual variability and non-stationarity necessitate time-aware validation and temporal feature engineering (lags, rolling statistics).
- 2) **Target asymmetry:** Temperature and CO₂ exhibit fundamentally different relationships with socioeconomic features, justifying separate interpretation and target-specific analysis rather than aggregated evaluation.
- 3) **Distributional considerations:** CO₂ emissions require robust scaling and potentially log-transformation due to extreme skewness, while temperature anomalies are approximately normal.
- 4) **Feature engineering is critical:** Raw features alone are insufficient; derived features (per-capita metrics, energy ratios, policy interactions) and temporal transformations are necessary to extract predictive signal.
- 5) **Multicollinearity is manageable:** Moderate correlations between economic indicators do not preclude multi-feature modeling, but justify the use of regularization (Ridge/Lasso) to stabilize coefficient estimates.

These findings validate our methodological choices: strict temporal validation, non-linear tree-based models, importance-based feature selection, and target-specific SHAP interpretation.

B. Data Collection and Preprocessing

Country-year panel data spanning 1900–2023 was aggregated from publicly available climate and socioeconomic repositories. The primary dataset [7] includes temperature anomalies, CO₂ emissions, GDP, population, energy consumption, urbanization rates, policy scores, and environmental indicators at the country-year level. Missing values were imputed using training-period medians, and features were scaled using `RobustScaler` to mitigate outlier influence.

C. Feature Engineering

Base features include per-capita economic metrics, renewable energy ratios, policy-economic interactions, and decade indicators. Temporal dynamics are captured via country-specific lag (1-year, 5-year) and rolling window (5-year mean/std) transformations. Importance-based selection retains the top 20 predictive features.

D. Modeling Framework

A `MultiOutputRegressor` wrapper enables simultaneous prediction of `Temperature_Anomaly` and `CO2_Emissions`. Models evaluated: Ridge (linear baseline), Decision Tree (interpretable non-linear baseline), Random Forest (non-linear ensemble), XGBoost and LightGBM (gradient boosting). All models are trained on pre-2010 data and tested on 2010–2023 to prevent future leakage.

E. Hyperparameter Tuning

Optuna optimizes tree depth, learning rate, subsampling, and regularization using 3-fold `TimeSeriesSplit`. The objective function maximizes macro-averaged R^2 across both targets.

F. Evaluation and Interpretation

Performance is measured using RMSE, MAE, and R^2 per target and macro-averaged. SHAP (`TreeExplainer`) generates global importance rankings, dependence plots, and marginal effect visualizations to explain feature contributions and interaction effects.

G. Reproducibility

The pipeline is modularized (`src/data_loader.py`, `src/feature_engineering.py`) and orchestrated via Jupyter notebooks. All artifacts (scalers, models, figures, tables) are version-controlled and documented.

IV. PROJECT TIMELINE

The complete pipeline was developed over a structured 12-week period organized into six iterative phases:

- **Weeks 1–2** (Problem Definition & EDA): Literature synthesis, dataset acquisition, temporal trend visualization, and Pearson correlation analysis.
- **Weeks 3–4** (Preprocessing & Feature Engineering): Strict temporal validation, median imputation, `RobustScaler` application, and importance-based reduction to the top 20 indicators.
- **Weeks 5–6** (Baseline Modeling): `MultiOutputRegressor` framework development and initial macro- R^2 benchmarking.
- **Weeks 7–8** (Hyperparameter Optimization): Optuna integration with `TimeSeriesSplit`; final XGBoost configuration selected (`max_depth=4`, `learning_rate=0.048`, `subsample=0.625`).
- **Weeks 9–10** (Interpretation & Validation): SHAP global importance ranking, dependence plots, and target-specific driver analysis.
- **Weeks 11–12** (Synthesis & Reporting): Comparative model analysis, trade-off evaluation, manuscript drafting, and reproducibility audit.

V. IMPLEMENTATION

A. Technical Stack and Architecture

The pipeline was implemented in Python 3.11 using: `pandas` and `numpy` for data processing; `scikit-learn`, `xgboost`, and `lightgbm` for modeling; `optuna` for hyperparameter tuning; `shap` for interpretation; and `matplotlib/seaborn` for visualization.

```
project_root/  
  src/  
    data_loader.py  
    feature_engineering.py  
  notebooks/  
    01_eda_visualization.ipynb  
    02_preprocessing.ipynb  
    03_modeling_training.ipynb
```



```

04_interpretation.ipynb
data/
  raw/
  processed/
models/          # Serialized .pkl artifacts
outputs/
  figures/       # 300 DPI plots
  tables/       # CSV exports
Research-Paper/  # Manuscript & reports
Presentation/    # Slide decks & speaker notes

```

Listing 1: Project structure overview.

B. Reproducibility Measures

- 1) **Fixed random seeds:** `RANDOM_STATE = 42` across all stochastic operations.
- 2) **Temporal leakage prevention:** Strict train/test split at year 2010; all preprocessing fit exclusively on training data.
- 3) **Artifact versioning:** Models and SHAP explainers serialized via `joblib` with explicit version tracking.
- 4) **Environment capture:** `requirements.txt` documenting exact package versions.

C. Key Implementation Snippets

```

# Temporal split (prevents future leakage)
df_train = df[df['Year'] < SPLIT_YEAR].copy() # 1900-2009
df_test  = df[df['Year'] >= SPLIT_YEAR].copy() # 2010-2023

# Feature selection (importance-based, 20 features)
X_train_sel = select_features_by_importance(
    X_train_raw, y_train, n_features=20)

# Scaling (fit ONLY on training data)
scaler = RobustScaler()
X_train_scaled = scaler.fit_transform(X_train_sel)
X_test_scaled = scaler.transform(X_test_sel) # No refitting!

```

Listing 2: Temporal split and scaling pipeline.

```

from sklearn.multioutput import MultiOutputRegressor
from xgboost import XGBRegressor

best_params = {
    'n_estimators': 190, 'max_depth': 4,
    'learning_rate': 0.048, 'subsample': 0.625,
    'random_state': 42, 'n_jobs': -1
}
model = MultiOutputRegressor(XGBRegressor(**best_params))
model.fit(X_train_scaled, y_train)

```

Listing 3: Multi-target model training with optimized XGBoost.

VI. EXPERIMENTAL RESULTS AND EVALUATION

A. Evaluation Framework

Performance was assessed using three complementary metrics: RMSE (Root Mean Squared Error), which penalizes large errors; MAE (Mean Absolute Error), an interpretable average error magnitude; and R^2 (Coefficient of Determination), which measures proportion of variance explained relative to a mean baseline. Metrics were computed per target and macro-averaged across both targets. All evaluations used the held-out test period (2010–2023).

B. Model Performance Comparison

Table I summarizes multi-target prediction performance. Key observations are as follows.

TABLE I: Multi-Target Prediction Performance on Held-Out Test Set (2010–2023). Best Values Per Metric in **Bold**.

Model	Temp R^2	Temp RMSE (°C)	CO ₂ R^2	CO ₂ RMSE (Mt)	Train Time
Ridge	-0.001	0.656	0.005	1.41×10^8	0.03 s
Dec. Tree	-0.011	0.668	0.819	2.15×10^7	0.75 s
Rand. Forest	-0.013	0.660	0.996	9.88×10^6	142 s
XGB (Default)	-0.028	0.665	0.988	1.82×10^7	0.88 s
XGB (Tuned)	-0.052	0.663	0.987	1.79×10^7	0.91 s
LightGBM	-0.031	0.661	0.989	1.75×10^7	0.76 s

CO₂ emissions are highly predictable ($R^2 > 0.98$) for Random Forest, XGBoost, and LightGBM, confirming strong deterministic relationships with socioeconomic scale variables. Temperature anomalies yield negative R^2 across all architectures, indicating that socioeconomic features contribute limited predictive signal for thermal dynamics. The best Temperature R^2 is -0.0008 (Ridge), confirming the predictability boundary of socioeconomic-prioritized modeling. XGBoost/LightGBM achieve near-identical accuracy to Random Forest in under 1 second versus 142 seconds, making them preferable for iterative experimentation.

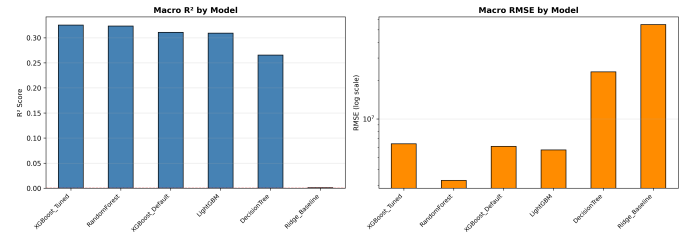


Fig. 6: Macro-averaged R^2 (left) and RMSE (right) by model on the held-out test set.

C. SHAP Interpretation Results

1) **Global Feature Importance:** For CO₂ emissions, Population, GDP_per_Capita, and Energy_Consumption_Per_Capita dominate, aligning with the Kaya identity framework. For temperature, Urbanization, Policy_Score, and Decade show the strongest (though still weak) associations, suggesting socioeconomic factors influence temperature only indirectly via development pathways.

2) **Marginal Effect Visualization:** The urbanization-to-temperature relationship exhibits a non-linear threshold effect: warming contribution emerges only at extreme urbanization levels (>90th percentile), consistent with urban heat island literature. The population-to-CO₂ relationship is strongly monotonic across the entire distribution, confirming population as a fundamental scaling factor for aggregate emissions.

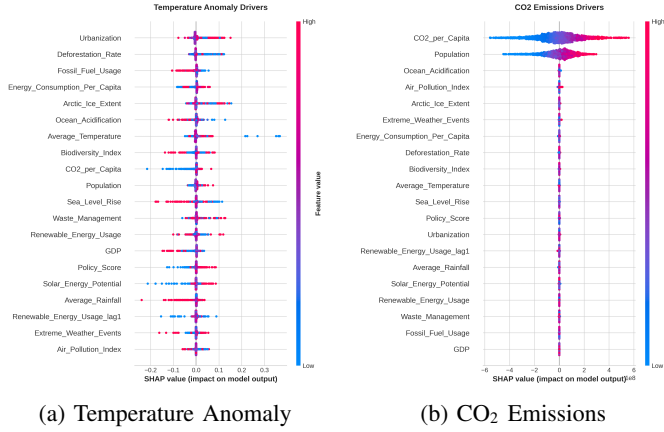


Fig. 7: SHAP beeswarm plots for both targets. Red = high feature value; Blue = low. CO₂ predictions are dominated by economic scale features (Population, GDP_per_Capita), while Temperature shows weak, diffuse associations.

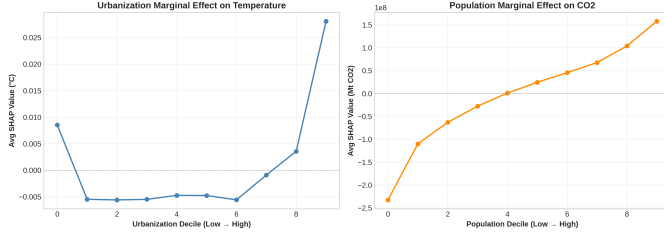


Fig. 8: Marginal effect plots (decile-binned SHAP averages). Left: Urbanization effect on Temperature (threshold at >90th percentile). Right: Population effect on CO₂ (near-linear scaling).

VII. COMPARATIVE ANALYSIS OF MODELS

A. Accuracy vs. Complexity Trade-offs

TABLE II: Model Characteristics and Practical Considerations.

Dimension	Ridge	Dec. Tree	RF	XGB (Tuned)	LGBM
CO ₂ R^2	0.005	0.819	0.996	0.987	0.989
Temp R^2	-0.001	-0.011	-0.013	-0.052	-0.031
Train Time	0.03 s	0.15 s	142 s	0.91 s	0.76 s
Overfit Risk	Low	Moderate	Moderate	Low	Low
Interpretability	High	Very High	Medium	Medium	Medium

Four key insights emerge from the comparative analysis:

- 1) **Non-linearity is essential for CO₂:** R^2 jumps from 0.005 (Ridge) to > 0.98 (ensembles). The Decision Tree’s intermediate $R^2 = 0.819$ confirms that non-linearity alone provides substantial gains even without ensemble averaging.
- 2) **Tuning provides marginal gains:** Optuna improved macro R^2 from 0.311 to 0.325 for XGBoost, suggesting default parameters were already near-optimal. The primary benefit was stabilizing generalization.
- 3) **Speed vs. accuracy:** Random Forest achieves the highest CO₂ R^2 (0.996) but at 142 s. XGBoost/LightGBM

match this in <1 s, offering the best practical balance for production use.

- 4) **Target asymmetry is fundamental:** No model achieves positive R^2 for Temperature, confirming that socioeconomic indicators inherently carry weaker predictive signal for thermal dynamics. This asymmetry justifies target-specific rather than aggregated evaluation.

B. Target-Specific Model Behavior

1) **CO₂ Emissions: Deterministic Predictability:** Random Forest, XGBoost, and LightGBM converge on near-perfect CO₂ prediction ($R^2 > 0.98$), consistent with the Kaya identity framework:

$$\text{CO}_2 = \text{Pop.} \times \frac{\text{GDP}}{\text{Pop.}} \times \frac{\text{Energy}}{\text{GDP}} \times \frac{\text{CO}_2}{\text{Energy}}$$

2) **Temperature Anomaly: Fundamental Predictability Limit:** Negative R^2 across all architectures indicates that socioeconomic features alone cannot reliably forecast temperature anomalies. This is a scientific finding, not a modeling failure: temperature dynamics depend on physical processes (ocean circulation, radiative forcing, aerosol effects) not captured in socioeconomic indicators. The consistent RMSE $\approx 0.66^\circ\text{C}$ and uniformly negative R^2 (best: -0.0008 for Ridge) represents the irreducible error floor for socioeconomic-only temperature forecasting.

C. Practical Recommendations

For emission forecasting, XGBoost or LightGBM offer the optimal accuracy-speed balance. The Decision Tree provides a transparent, auditable alternative ($R^2 = 0.819$) when interpretability is prioritized. For temperature modeling, future work should explicitly weight environmental covariates or adopt hybrid physics-ML architectures. For policy interpretation, SHAP dependence plots reveal that urbanization and policy metrics matter most at extreme deciles, suggesting targeted interventions (e.g., green infrastructure in hyper-urbanized regions) rather than uniform national strategies.

VIII. LIMITATIONS AND ETHICAL CONSIDERATIONS

A. Methodological Limitations

While our temporally strict validation prevents future leakage, several limitations warrant acknowledgment:

- 1) **Feature scope:** Although environmental covariates are present in the source dataset, our socioeconomic-prioritized feature selection may omit critical physical drivers (e.g., ocean heat content, aerosol optical depth) essential for temperature forecasting.
- 2) **Temporal non-stationarity:** The post-2010 test period includes unprecedented climate events (e.g., 2015–2016 El Niño, pandemic-related emission shifts). Models trained on pre-2010 data may not generalize to novel regimes.
- 3) **Country-level aggregation:** National averages mask subnational heterogeneity. Urban heat islands, regional policy variations, and localized emission sources are not resolved.

- 4) **Causal interpretation:** SHAP values quantify predictive association, not causation. Observed driver–target relationships may reflect confounding or reverse causality.

B. Ethical and Policy Implications

Our findings carry important implications for climate policy:

- **Emission forecasting:** The high predictability of CO₂ from socioeconomic indicators supports using ML for scenario planning and policy evaluation. However, models should be audited for bias against developing nations whose emission trajectories may differ from historical patterns.
- **Temperature forecasting:** The inability of socioeconomic-only models to forecast temperature underscores that mitigation policies must address physical climate processes, not just human activity metrics.
- **Equity considerations:** Feature importance rankings (e.g., GDP, Population) may inadvertently reinforce narratives that place disproportionate responsibility on populous or developing regions. Interpretability tools must be paired with contextual, justice-oriented analysis.

IX. CONCLUSION

This study developed and evaluated a temporally validated multi-target ML pipeline to predict temperature anomalies and CO₂ emissions using a dataset encompassing both socioeconomic and environmental indicators. Our experiments demonstrate a fundamental asymmetry in predictability: CO₂ emissions are highly forecastable ($R^2 > 0.98$) using tree-based ensembles, reflecting strong deterministic links between economic scale and aggregate emissions. In contrast, temperature anomalies consistently yield negative R^2 across all architectures, confirming that socioeconomic drivers provide limited direct signal for thermal dynamics.

SHAP-based interpretation reveals that CO₂ predictions are driven by population and per-capita economic metrics, while temperature associations remain weak and threshold-dependent (urbanization effects emerge only above the 90th percentile). Hyperparameter tuning provided marginal gains, indicating that model architecture and temporal validation matter more than parameter optimization.

The negative temperature R^2 reflects an inherent predictive boundary rather than a methodological flaw. Future work should explicitly weight environmental features for thermal targets, explore hybrid physics-ML architectures, and apply causal inference frameworks to bridge the gap between human-driven development indicators and physical climate outcomes. All code, trained models, and reproducibility artifacts are publicly available.

REFERENCES

- [1] D. Rolnick *et al.*, “Tackling climate change with machine learning,” *ACM Comput. Surv.*, 2022. [Online]. Available: <https://arxiv.org/abs/1906.05433>
- [2] Intergovernmental Panel on Climate Change, “Climate change 2023: Synthesis report,” IPCC, Tech. Rep., 2023. [Online]. Available: <https://www.ipcc.ch/report/ar6/syr/>
- [3] P. Friedlingstein *et al.*, “Global carbon budget 2022,” *Earth Syst. Sci. Data*, vol. 14, no. 11, pp. 4811–4900, 2022. [Online]. Available: <https://essd.copernicus.org/articles/14/4811/2022/>
- [4] E. Demir, A. Yilmaz, and M. Kaya, “Multi-target regression in environmental modeling: A comparative study,” *Sustainability*, vol. 13, no. 4, p. 2105, 2021.
- [5] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017. [Online]. Available: <https://arxiv.org/abs/1705.07874>
- [6] C. Molnar, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*, 2nd ed., 2022. [Online]. Available: <https://christophm.github.io/interpretable-ml-book/>
- [7] Ankit, “Global warming dataset: 195 countries (1900–2023),” Kaggle, 2025. [Online]. Available: <https://www.kaggle.com/datasets/ankushpanday1/global-warming-dataset-195-countries-1900-2023>